TeraMove-H — Walkthrough

Plain-language guide to the paper, with every formula
(main focus: user association)

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1 The idea

THz access points (APs) need line of sight, and the people they serve are also what blocks them. TeraMove fixed blockage by *moving* APs. TeraMove-H tries two cheaper fixes first:

1. a **Wi-Fi layer** together with smarter **user association**;
2. **power control**: idle APs are switched off, and busy APs use less power.

APs move only if these electronic fixes are not enough.

2 Setup

Building	$32 \times 20 \times 3$ m floor, 12 rooms
THz APs	14 APs from the MILP (Fig. 1); state $\theta_a = (x_a, y_a, z_a, \phi_a, \alpha_a)$
AP height	$z_a \in \{2.0, 2.2, \ldots, 3.0\}$ m
AP azimuth	ϕ_a in 45° steps, compass bearing (0° = North, clockwise)
Beam width	$\alpha_a \in \{30, 45, 60, 75, 90, 120\}°$
AP movement	along a ceiling rail inside its own room only
Wi-Fi APs	4 fixed ceiling APs, 6 GHz, 160 MHz each, separate channels (no mutual interference)
People	80 % in 6 Gaussian clusters ($\sigma = 1$ m), 20 % uniform; bodies never overlap
Body model	cylinder, radius 0.2 m, height 1.6 m
Users	each person is an active user with probability 0.5; device at 1.0 m, random facing direction
Demand	every user needs at least $R_{\min}$ (default 1 Gb/s)

3 Link model

3.1 THz path loss (dB)

Path loss

$$L_{a,u} = \underbrace{20 \log_{10} \frac{4\pi f d}{c}}_{\text{free space}} + \underbrace{4.343\, k(f)\, d}_{\text{absorption}} + \underbrace{L^{\text{st}}_{a,u}}_{\text{walls}} + \underbrace{\min\left(n_{a,u} L_h,\ L^{\max}_h\right)}_{\text{bodies}} + \underbrace{L_{\text{sb}}\, \mathbf{1}^{\text{sb}}_{a,u}}_{\text{own body}}$$

Why: Each term is a separate reason the signal gets weaker, so the losses (in dB) simply add up. *Free space*: the wave spreads over a growing sphere, so power falls with d^2; at 300 GHz this is already about 82 dB at 1 m. *Absorption*: water vapour in the air soaks up THz energy, a fixed number of dB per metre. *Walls*: each wall crossed costs a fixed amount that depends on its material. *Bodies*: a person in the path costs about 20 dB; we cap the total at 40 dB because after two bodies the link is dead anyway. *Own body*: if the AP is behind you, your own body blocks it.

Symbol	Meaning	Value
f	carrier frequency	300 GHz
d	3-D AP–user distance	–
c	speed of light	–
$k(f)$	molecular absorption (ITU-R P.676)	$1.54 \times 10^{-3}\,\mathrm{m}^{-1}$ (6.68 dB/km)
L^{st}	wall penalty per wall crossed	concrete 30, wood 20, glass 15 dB
$n_{a,u}$	bodies whose circle the ray crosses at height ≤ 1.6 m	–
$L_h,\ L_h^{\max}$	body loss, cap	20 dB, 40 dB
$\mathbf{1}^{\mathrm{sb}}$	AP lies behind the user's own body (3GPP self-blocking region)	0 or 1
L_{sb}	self-blocking loss	30 dB

Linear path gain:

$$\ell_{a,u} = 10^{-L_{a,u}/10}.$$

Why: Powers from different APs must be *added*, and adding only works on linear values, not on dB.

3.2 AP antenna gain (energy conserving)

The main lobe covers the lower half-space within $\pm\alpha/2$ of ϕ_a. Its solid angle is α steradians, so

AP gain

$$G_m(\alpha) = \frac{4\pi - g_s\,(4\pi - \alpha)}{\alpha}, \qquad g_s = -10 \text{ dBi (side lobe)}, \quad \alpha \text{ in radians.}$$

Why: An antenna cannot create energy; it only redirects it. Averaged over all directions the gain must be 1 (i.e. 4π when integrated), so $G_m \cdot \alpha + g_s(4\pi - \alpha) = 4\pi$. Solving for G_m gives the formula: squeeze the beam (small α) and the main gain goes up.

A wider beam gives less gain: **13.4 / 10.4 / 7.4 dBi** for $\alpha = 30° / 60° / 120°$. The AP gain towards user u is

$$G^t_{a,u} = \begin{cases} G_m(\alpha_a), & \Delta\phi_{a,u} \leq \alpha_a/2, \\ g_s, & \text{otherwise.} \end{cases}$$

Why: A user inside the sector gets the full beam; outside it, only the weak side-lobe leakage.

3.3 User device antenna

The device points a $G_r = 20$ dBi pencil beam at its serving AP. The same energy rule fixes the main-lobe cone, with half-angle ψ:

$$\Omega_r = \frac{4\pi(1 - g_r)}{G_r - g_r}, \qquad \psi = \arccos\!\Big(1 - \frac{\Omega_r}{2\pi}\Big) = 10.9°.$$

Why: Same energy rule for the phone's pencil beam. A cone of half-angle ψ covers $2\pi(1 - \cos\psi)$ steradians, which gives ψ. Only APs inside this narrow cone can interfere strongly with the user.

$G^r_{u,a,b}$ is the device gain towards AP b when the beam points at AP a. It equals G_r if b lies within ψ of the direction to a, and $g_r = -10$ dBi otherwise.

3.4 SINR and capacity

Write $h_{b,u} = G^t_{b,u}\,\ell_{b,u}$ for the channel gain.

SINR, noise and capacity

$$\mathrm{SINR}_{u,a} = \frac{p_a\,h_{a,u}\,G_r}{N_0 + \sum_{b\neq a} p_b\,h_{b,u}\,G^r_{u,a,b}}$$

$$N_0 = -174 + 10\log_{10} B + \mathrm{NF} = -174 + 100 + 10 = -64 \text{ dBm}$$

$$c_{u,a} = \begin{cases} B\,\min\big(\log_2(1+\mathrm{SINR}_{u,a}),\,6\big), & \mathrm{SINR}_{u,a} \geq \gamma_{\mathrm{th}} = 5 \text{ dB}, \\ 0, & \text{otherwise}, \end{cases} \qquad B = 10 \text{ GHz}.$$

Why: *SINR*: useful power divided by everything unwanted (noise plus the power of all other APs at this user). *Noise*: thermal noise is −174 dBm per Hz, so a wide 10 GHz channel collects +100 dB of it, plus 10 dB added by the receiver. *Capacity*: Shannon's formula gives bits/s/Hz $= \log_2(1 + \mathrm{SINR})$; we cap it at 6 (64-QAM, the best practical modulation) and set it to 0 below 5 dB, where a THz link cannot be held.

3.5 Wi-Fi link

3GPP InH-Office path loss, with $f_c = 6$ GHz:

$$\mathrm{PL}_{\mathrm{LOS}} = 32.4 + 17.3\log_{10} d + 20\log_{10} f_c,$$

$$\mathrm{PL}_{\mathrm{NLOS}} = \max\big(\mathrm{PL}_{\mathrm{LOS}},\; 17.3 + 38.3\log_{10} d + 24.9\log_{10} f_c\big).$$

Why: These are standard 3GPP curves fitted to indoor measurements. The distance exponent is 1.73 in line of sight and 3.83 when walls or people are in the way. The max makes sure a blocked link is never better than a clear one.

A link is LOS only if no full-height wall and no body is in the way. The rate is the 2-stream 160 MHz PHY rate of the highest 802.11ax MCS whose sensitivity is met, times a MAC efficiency of 0.7. The maximum Wi-Fi rate is about **1.68 Gb/s**.

4 User association (the core)

4.1 Airtime

Every AP time-shares its channel. If user u is served by AP a, it needs this share of the AP's time:

Airtime needed

$$\tau_{u,a} = \frac{R_{\min}}{c_{u,a}}$$

Why: To deliver $R_{\min}$ over a link of speed c, the AP must spend a fraction $R_{\min}/c$ of its time on that user. Example: 1 Gb/s on a 40 Gb/s link needs 2.5 % of the time.

Each AP can give out at most **1** unit of time in total.

Link	Capacity	Airtime needed at 1 Gb/s
Good THz	40 Gb/s	0.025
Good Wi-Fi	1.5 Gb/s	0.67 (two-thirds of the whole AP)

4.2 Optimisation problem

Let $x_{u,a} = 1$ if user u is served by AP a, and 0 otherwise.

Association problem (maximum GAP)

$$\begin{aligned}
\max_{\mathbf{x}} \quad & \sum_u \sum_a x_{u,a} & & \\
\text{s.t.} \quad & \sum_a x_{u,a} \le 1 & \forall u \quad & \text{(at most one AP per user)} \\
& \sum_u \tau_{u,a}\, x_{u,a} \le 1 & \forall a \quad & \text{(airtime budget)} \\
& x_{u,a} = 0 \ \text{ if } c_{u,a} < R_{\min} & & \text{(link too weak)} \\
& x_{u,a} \in \{0,1\} & &
\end{aligned}$$

Why: We count served users. Each user takes one AP at most; an AP cannot hand out more than 100 % of its time; and a link slower than $R_{\min}$ is useless even with all the time.

Satisfaction is the fraction of users who get at least $R_{\min}$:

$$S = \frac{1}{|\mathcal{U}|} \sum_u \sum_a x_{u,a}.$$

Why: Satisfaction is simply the share of users who got what they asked for.

This is a **maximum Generalised Assignment Problem (GAP)** in which every user has the same value. Think of packing items (users) into bins (APs), where an item's size depends on the bin it goes into.

4.3 Why it is hard (Proposition 1)

Deciding whether *all* users can be served is NP-complete. It contains the PARTITION problem: given integers $s_1, \ldots, s_k$ with $\sum_i s_i = 2T$, can they be split into two halves of equal sum? Use two APs and choose the capacities so that

$$c_{u,1} = c_{u,2} = \frac{R_{\min}\, T}{s_u} \quad \Longrightarrow \quad \tau_{u,a} = \frac{s_u}{T}.$$

Both airtime budgets hold with everyone served **if and only if** the integers split into two halves that each sum to T. So an exact solver is too slow for every evaluation, and we need a fast algorithm: the simple greedy (baseline) and **PACE** (proposed).

Why: If an exact solver could always answer quickly, it would also solve PARTITION quickly, which is believed impossible. So we need a fast algorithm that is still close to optimal, and ideally one that can tell us when it *is* optimal.

4.4 Greedy algorithm (`gap_greedy`)

1. List every usable (user, AP) pair, THz and Wi-Fi together.
2. Sort the pairs by airtime $\tau_{u,a}$, smallest first.
3. Walk down the list. Assign a pair only if the user is still free **and** the AP still has enough time left.
4. Users left over are unserved.

The cost is $O(KM \log KM)$ for K users and M APs. Cheapest pairs first leaves room for the most users. With a single AP this rule is exactly optimal.

Why: Short tasks first: giving time to the users who need the least of it leaves the most time for others, like packing small items into a suitcase first. Because THz and Wi-Fi pairs are in one list, a blocked user automatically falls back to Wi-Fi, and a full THz AP pushes users to a second-best AP.

Worked example ($R_{\min} = 1$ Gb/s).

User	THz-A	Wi-Fi
u1	20 Gb/s $\to \tau = 0.05$	1.6 Gb/s $\to \tau = 0.63$
u2	4 Gb/s $\to \tau = 0.25$	1.2 Gb/s $\to \tau = 0.83$
u3	blocked	1.5 Gb/s $\to \tau = 0.67$

Sorted pairs: (u1,A) 0.05 $\to$ (u2,A) 0.25 $\to$ (u1,W) 0.63 $\to$ (u3,W) 0.67 $\to$ (u2,W) 0.83.

Step	Pair	Action	Load A	Load W
1	(u1, A)	assign	0.05	0
2	(u2, A)	assign	0.30	0
3	(u1, W)	skip (u1 already served)	0.30	0
4	(u3, W)	assign	0.30	0.67
5	(u2, W)	skip (u2 already served)	0.30	0.67

All 3 users are served, and Wi-Fi is spent only on the blocked user.

4.5 Where the greedy goes wrong

The greedy looks only at airtime. A user who has *other options* can take the last airtime of an AP that another user *cannot do without*.

User	AP A	AP B
u1	$\tau = 0.4$	$\tau = 0.5$
u2	$\tau = 0.7$	blocked

- **Greedy:** (u1, A) is the cheapest pair, so u1 takes A (load 0.4). u2 then needs $0.4 + 0.7 > 1$ on A and is lost. **1 user served.**
- **Optimum:** u1 on B, u2 on A. **2 users served.**

Why: Greedy decides pair by pair and never looks back. It cannot see that A is the only option for u2. In real instances this happens on crowded Wi-Fi APs and on THz APs at the edge of a crowd, mostly at low demand when each AP packs many users.

4.6 PACE: the proposed association algorithm

PACE = **P**rice-guided **A**ssociation with **C**ertified **E**jection chains. It starts from the greedy solution and adds four ideas. Code: function `gap_pace` in `teramove_h/association.py`.

Part 1 — Slot cuts: how many users can an AP hold at most?

Sort AP a's airtimes from small to large. K_a is the largest n whose n smallest airtimes still fit:

Slot cut — Proposition 2 in the paper

$$\tau_{(1),a} + \tau_{(2),a} + \cdots + \tau_{(n),a} \le 1 \quad \Longrightarrow \quad \sum_u x_{u,a} \le K_a \qquad \forall a.$$

Why: Any n users on AP a need at least the n smallest airtimes. If those already exceed 1, n users cannot fit. The fractional (LP) version of the problem ignores this: at 1 Gb/s a Wi-Fi AP (at most 1.68 Gb/s, so $\tau \ge 0.6$) really holds *one* user, but the fractional budget "holds" 1.68 users. The cut removes this fake capacity, so the bound in Part 2 becomes tight.

Part 2 — Prices and an upper bound (Lagrangian dual)

Give each AP an **airtime price** $\lambda_a \geq 0$ and a **slot price** $\nu_a \geq 0$. For any prices define

Dual function

$$L(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \sum_a (\lambda_a + K_a \nu_a) + \sum_u \max\Big(0,\ \max_a (1 - \lambda_a \tau_{u,a} - \nu_a)\Big).$$

Why: Read it as a market. Each user is worth 1. Joining AP a costs $\lambda_a \tau_{u,a} + \nu_a$. Each user picks its cheapest AP and joins only if the cost is below 1. A crowded AP gets a high price, which pushes flexible users elsewhere.

Upper bound (weak duality) — Proposition 3 in the paper

For *any* prices $\boldsymbol{\lambda}, \boldsymbol{\nu} \geq 0$:

$$\text{maximum number of users served} \ \leq\ \lfloor L(\boldsymbol{\lambda}, \boldsymbol{\nu}) \rfloor.$$

Why: Take any valid assignment and add the non-negative leftovers $\lambda_a(1 - \sum_u \tau_{u,a} x_{u,a})$ and $\nu_a(K_a - \sum_u x_{u,a})$ to the user count; the count can only grow. Rearranged, each user contributes at most its best "value minus price", which is the max term in L. The count is an integer, so take the floor.

Optimality certificate

If PACE serves n users and finds prices with $L < n + 1$, then n is **provably optimal**.

Why: No assignment can serve more than $\lfloor L \rfloor = n$ users, so n is the best possible, and no exact solver is needed to know it. The best possible L equals the LP relaxation *with* slot cuts (Geoffrion 1974), and with the cuts it is very tight, so the certificate fires often.

Updating the prices (subgradient step). Let $u \to a$ mean "users whose cheapest AP is a". Measure how far each AP is from its limits, then move the prices against that gap:

Price update

$$g_a^{\lambda} = 1 - \sum_{u \to a} \tau_{u,a}, \qquad g_a^{\nu} = K_a - \big|\{u \to a\}\big|,$$

$$(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leftarrow \max\Big(0,\ (\boldsymbol{\lambda}, \boldsymbol{\nu}) - \vartheta \, \frac{L - (n + \frac{1}{2})}{\|\mathbf{g}\|^2} \, \mathbf{g}\Big).$$

Why: This is supply and demand. An over-booked AP ($g < 0$) gets a *higher* price, and an AP with spare time ($g > 0$) gets a lower one. The step size shrinks as L approaches the target $n + \frac{1}{2}$ (Polyak's rule, Held–Wolfe–Crowder 1974). ϑ starts at 1 and halves after 20 steps without progress. Prices never go below 0.

Part 3 — Price-guided restarts

Every 5 price steps, PACE re-runs the greedy, but sorts the pairs by the **priced cost** instead of the raw airtime (ties by airtime):

$$\text{sort key} = \lambda_a \tau_{u,a} + \nu_a.$$

It keeps the result if it serves more users.

Why: With all prices at 0 this is exactly the old greedy. Nonzero prices tell greedy which APs are scarce, so each price vector gives a different, smarter order.

Part 4 — Ejection chains (push someone along)

An unserved user u can get in by moving other users:

- **depth 1:** u enters AP a by displacing v; v moves to an AP b with free time;
- **depth 2:** v instead displaces w at b, and w moves to a third AP c.

Why: Each chain serves *one more* user and drops nobody. It is the assignment-problem version of the "augmenting path" in matching (Yagiura, Ibaraki and Glover 2004). In the example above, one depth-1 chain fixes the greedy: u2 enters A, and u1 moves to B.

The full algorithm

1. Run greedy, then depth-1 chains. This serves n users.
2. Repeat up to 200 times:
   - compute L; if the best $L < n + 1$, **stop** (optimal, certified);
   - every 5 steps, do a price-guided restart and keep it if it serves more users;
   - take one price step.
3. If no certificate was found, try depth-2 chains.
4. Move served users to an AP with smaller τ where it fits. The count stays the same, and less airtime is used.

Each price step costs $O(KM)$ for K users and M APs; the median run time is about 0.6 ms.

Guarantees

- **Never worse than greedy:** PACE starts from the greedy solution and keeps a new one only if it serves strictly more users.
- **Always feasible:** every move checks the airtime budget with the same tolerance as the greedy.
- **Known gap:** $\lfloor L \rfloor$ is always a valid upper bound, even without a certificate.

Can it guarantee that every user is served? No algorithm can when capacity is short. PACE instead *proves* when it is impossible: if $\lfloor L \rfloor < k$, no association can serve all k users. TeraMove-H uses this at T0: when the bound shows that the target η is unreachable at the current geometry, moving APs is truly necessary, not a failure of the association algorithm.

Where the parts come from. Lagrangian relaxation (Geoffrion 1974; Fisher, Jaikumar and Van Wassenhove 1986), subgradient steps (Held, Wolfe and Crowder 1974), ejection chains (Yagiura, Ibaraki and Glover 2004) and pricing-based association (Shen and Yu 2014). Our contribution is to combine them for THz/Wi-Fi airtime association, together with the slot cuts and a certificate that decides whether APs must move.

4.7 Baselines

Rule	How it decides	Weakness
Max-SINR (old TeraMove)	strongest THz AP; no Wi-Fi	blocked users are lost; ignores load
Threshold offload	strongest THz AP if SINR ≥ 5 dB, else best Wi-Fi	ignores load; cannot pick a second-best THz AP
Greedy GAP (ablation)	cheapest airtime first, over all APs	blind to scarce APs (see above)
PACE (TeraMove-H)	prices, restarts and ejection chains, from the greedy start	slower, but below 1 ms median

4.8 How good are greedy and PACE?

Both were compared with the exact integer program, solved by HiGHS (`scipy.optimize.milp`), on **1,800** instances: 4 crowd sizes × 5 demands × 3 AP configurations × 30 seeds.

Check	Greedy	PACE
Equals the exact optimum	84.1 %	**99.4 %**
Users short of the optimum (all instances)	483	**12**
Worst instance	7 users short (87.0 %)	2 users short (96.6 %)
Optimality proved by its own bound	—	82.3 % (never wrong)
Median time	0.27 ms	0.62 ms (exact ILP: 34 ms)

PACE serves more users than the greedy in 282 instances and fewer in none.

$R_{\min}$ (Gb/s)	0.1	0.25	0.5	1	2
Instances where greedy is not optimal	156	91	39	0	0
Instances where PACE is not optimal	10	1	0	0	0
Users short, greedy	320	123	40	0	0
Users short, PACE	11	1	0	0	0
PACE certified optimal (%)	56.4	70.6	84.7	100	100

Why: The greedy fails only at low demand, where each AP packs many users and the packing gets tight. At 1 and 2 Gb/s an AP holds only a few users, and greedy, PACE and the optimum all coincide. This is why the main results at 1 Gb/s do not change with PACE (84.4 % with both), while the benefit appears at 0.1–0.5 Gb/s.

4.9 When association matters (full system)

Same seeds, same crowds and same search; only the association algorithm differs. Each cell shows PACE / greedy, averaged over the four crowd sizes (60 to 240 people) and 30 seeds.

$R_{\min}$ (Gb/s)	0.1	0.25	0.5	1	2
T0 S (%)	94.2 / 93.2	73.2 / 72.8	60.8 / 60.3	48.7 / 48.7	41.7 / 41.7
Final S (%)	98.3 / 98.0	92.5 / 92.4	86.1 / 85.9	80.4 / 80.3	75.1 / 75.2
No movement (%)	80 / 78	33 / 30	20 / 19	2 / 2	0 / 0
Displacement (m)	3.8 / 4.5	15.5 / 16.1	19.1 / 18.7	22.3 / 22.4	25.8 / 26.1

Why: T0 uses the same AP positions for both, so its difference is purely the association algorithm: +1.0 pp at 0.1 Gb/s (+2.7 pp with 180 people). After T0 the GA moves APs and repairs most association mistakes, so the final gain is small: +0.3 pp at 0.1 Gb/s ($p = 0.01$, significant), and not significant at higher demand. The real benefit is *less movement*: at 0.1 Gb/s, 80 % instead of 78 % of snapshots need no movement, and displacement drops from 4.5 to 3.8 m (−16 %). The cost is run time: about 2.5× longer per snapshot at 0.1 Gb/s, because PACE runs inside every fitness evaluation.

Certified escalation. In 457 snapshots, T0 fell short of $\eta = 0.9$ and TeraMove-H escalated. In **98.7 %** of them, the PACE bound $\lfloor L \rfloor$ proves that *no* association at the starting geometry could reach η. So the movement was truly necessary, not caused by a weak association heuristic.

PACE versus threshold offload.

$R_{\min}$ (Gb/s)	0.1	0.25	0.5	1	2
PACE − offload (pp)	+1.8	+2.0	+1.5	+0.9	−0.1 (not significant)

- **Low demand.** Each Wi-Fi AP can carry many users, so sharing its time wisely pays off. At 0.1 Gb/s, re-association alone serves **94.2 %** of users, and **80 %** of crowd snapshots need no AP movement at all.
- **High demand.** One Wi-Fi AP tops out at 1.68 Gb/s, so it carries about one user at 1 Gb/s. There is little left to allocate, and the rule barely matters.

5 Power control

5.1 Sleep

Any THz AP with no users is switched off ($p_a = 0$), and users are re-associated. This repeats for up to 3 rounds. A round is kept only if S does not drop.

5.2 Turn spare airtime into lower power

AP a's airtime load is

$$L_a = \sum_{u \in \mathcal{U}_a} \tau_{u,a} \leq 1.$$

Why: L_a is how much of AP a's time is actually used. If $L_a < 1$, the AP has spare time.

If every user's time is stretched by $1/L_a$, the user still gets $R_{\min}$ with a lower capacity:

Target SINR

$$c_u^\star = c_{u,a}\, L_a \qquad \Longrightarrow \qquad \gamma_u = \max\!\left(\gamma_{\text{th}},\; 2^{\,c_u^\star / B} - 1\right)$$

Why: Spare time can be traded for power. Give each user $1/L_a$ times more time, and it can live with a slower link $c_u^\star = c_{u,a} L_a$ and still get $R_{\min}$. Inverting Shannon ($c = B \log_2(1+\gamma)$) gives the SINR needed for that speed; never below 5 dB so the link stays up.

5.3 Yates minimum-power iteration

Power update

$$I_a(\mathbf{p}) = \max_{u \in \mathcal{U}_a} \frac{\gamma_u \left(N_0 + \sum_{b \neq a} p_b\, h_{b,u}\, G^r_{u,a,b} \right)}{G_r\, h_{a,u}}$$

$$\mathbf{p} \;\leftarrow\; \min\!\left(\mathbf{I}(\mathbf{p}),\; P_{\max}\right), \qquad P_{\max} = 20 \text{ dBm}$$

Why: Each AP picks the smallest power that lets its *weakest* user reach its target, given the interference it sees now. Changing one AP's power changes interference for the others, so we repeat until nothing changes. The min with $P_{\max}$ keeps powers legal.

Proposition 4 (paper numbering). $\mathbf{I}$ is a *standard interference function*:

Property	Holds because
Positive	$N_0 > 0$
Monotone	all coupling terms are ≥ 0
Scalable: $\mu\,\mathbf{I}(\mathbf{p}) > \mathbf{I}(\mu\mathbf{p})$ for $\mu > 1$	the N_0 term does not scale with μ

At the current powers $\mathbf{p}^0$, every served user already meets its target, so $\mathbf{I}(\mathbf{p}^0) \leq \mathbf{p}^0$. Starting from there, the iteration only goes **down** and converges to the minimum power vector $\mathbf{p}^\star$. At $\mathbf{p}^\star$, every served user keeps $R_{\min}$ and no AP exceeds its airtime budget:

$$c_u \geq c_u^\star \implies \frac{R_{\min}}{c_u} \leq \frac{\tau_{u,a}}{L_a} \implies \sum_u \frac{R_{\min}}{c_u} \leq 1.$$

Why: Every user's new speed is at least the planned $c_u^\star$, so its time need is at most the planned (stretched) share, and these shares add up to at most 1. Hence nobody loses service.

Result: THz power drops from **1.40 W to 0.53 W (−62.1 %)**, with about 4 of the 14 APs switched off. Satisfaction is unchanged.

6 Escalation and search

The stages run in order and stop at the first one that reaches $S \geq \eta = 0.9$:

Stage	What it does	Physical cost
T0	PACE association over THz + Wi-Fi at full power	none
T1	sleep + re-associate + minimum power	none
T2	genetic algorithm (GA) over azimuth ϕ and width α (1 budget unit)	none
M	GA over height, position within the room and beams (3 units), then Hungarian matching per room	movement

One budget unit is $40 \times 40 = 1600$ fitness evaluations. Final power control always runs at the end.

6.1 Fitness used by the search

Fitness

$$F = 0.6\,S + 0.4\,Q + 0.5 \cdot \mathbf{1}[S \geq \eta] - \beta M - \gamma C_p$$

Why: S is the main goal. Q gives partial credit, so the search can tell *almost* served from hopeless and climb towards better layouts. The bonus rewards crossing the 90 % target. M punishes moving hardware, and C_p stops APs from bunching up.

$$Q = \frac{1}{|\mathcal{U}|} \sum_u \min\Big(1,\ \frac{\max_a c_{u,a}}{R_{\min}}\Big) \qquad \text{(smooth link quality)}$$

$$M = \frac{\sum_a \|(x_a, y_a, z_a) - (x_a^0, y_a^0, z_a^0)\|}{|\mathcal{T}| \cdot \text{building diagonal}}$$

Why: Total distance moved, divided by its largest possible value, so the penalty stays between 0 and 1.

$$C_p = \frac{1}{\binom{n}{2}} \sum_{i<j} \max\Big(0,\ 1 - \frac{d_{ij}}{d_{\min}}\Big), \qquad d_{\min} = 3 \text{ m},$$

Why: Every pair of APs closer than 3 m adds a penalty that grows as they get closer; averaged over all pairs.

with $\beta = 0.25$ and $\gamma = 0.3$ (TeraMove configuration D).

6.2 Hungarian matching

New AP states are matched to physical APs inside each room so that total movement is minimal:

$$\sigma^\star = \arg\min_{\sigma} \sum_{i} C_{i,\sigma(i)}, \qquad C_{ij} = \|\mathbf{r}_i - \mathbf{r}'_j\|.$$

Why: All APs are identical, so it does not matter *which* AP goes to a new spot. The Hungarian algorithm picks the pairing with the least total travel, in $O(n^3)$ time.

6.3 Commissioning

The MILP layout's beams and heights were re-tuned once for the empty floor; the positions did not change. Empty-floor coverage rose from **66.2 % to 100 %**.

7 Main results

30 seeds, $R_{\min} = 1$ Gb/s, 120 people.

Method	Users satisfied S	AP movement
Static (no adaptation)	44.8 %	0 m
Old TeraMove	50.9 %	11.8 m
Re-association only (T0)	51.6 %	0 m
T0–T2 (no movement)	58.9 %	0 m
THz only (no Wi-Fi)	77.2 %	25.8 m
Threshold offload	83.0 %	23.3 m
TeraMove-H with greedy association	84.4 %	24.6 m
TeraMove-H (PACE)	**84.4 %**	**24.7 m**
Flat GA (all dimensions at once)	87.3 %	30.4 m

At 1 Gb/s greedy is already optimal, so PACE and greedy give the same satisfaction here; PACE's gains appear at lower demand (Section 4). The flat GA satisfies 2.9 pp more users but moves APs about 23 % more. That is a trade-off, not a win for either method.

8 Where it lives in the code

File	What is in it
`teramove_h/association.py`	`gap_pace` (proposed), `gap_greedy`, `gap_exact`, `slot_limits`, `gap_lp_bound`, `max_sinr_pref`, `naive_offload_pref`
`teramove_h/evaluator.py`	association → sleep → power control pipeline
`teramove_h/power.py`	SINR targets γ_u and the Yates iteration
`teramove_h/network.py`	SINR and capacity for every (user, AP) pair
`teramove_h/antenna.py`	energy-conserving gains $G_m(\alpha)$ and ψ
`teramove_h/channel.py`	ray geometry, walls, bodies, self-blocking, Wi-Fi rates
`teramove_h/cascade.py`	escalation stages T0 → T1 → T2 → M, and every baseline
`paper/main.tex`	Sec. III (model), Sec. IV-B (association), Sec. V (results)